

Computer Architecture

Course Code: CSE 360

Presented by
Dr. Md. Nawab Yousuf Ali
Professor, Dept. of CSE

Lecture 1

Introduction

Objectives

After undergoing this course, students should be

- able to grasp the basic concept of computer architecture and organization and key skills of constructing cost-effective computer
- able to evaluate different designs and organizations of modern computers
- able to articulate design issues in the design and development of different components of a computer
- able to understand the design issues of scalable and parallel computer architectures

Scalable Computer:

It is the ability of a computer application or product (hardware or software) to continue to function well when it (or its context) is changed in size or volume to meet a user demand

Parallel Computer:

A **computer** that utilizes **parallel** processing that can be upgraded by adding more CPUs to it, effectively increasing its **computing** power.

Architecture & Organization

- **Computer architecture** refers to those attributes of a system visible to the programmer
 - Examples of architectural attributes includes:
 1. The instruction set,
 2. The number of bits used to represent various data types (e.g. numbers, characters)
 3. I/O mechanisms and
 4. Techniques for addressing memory

Architecture & Organization

- **Organizational attributes** include hardware details transparent to the programmer such as
 - 1.** Control signals
 - 2.** Interfaces between the computers and peripherals and
 - 3.** The memory technology

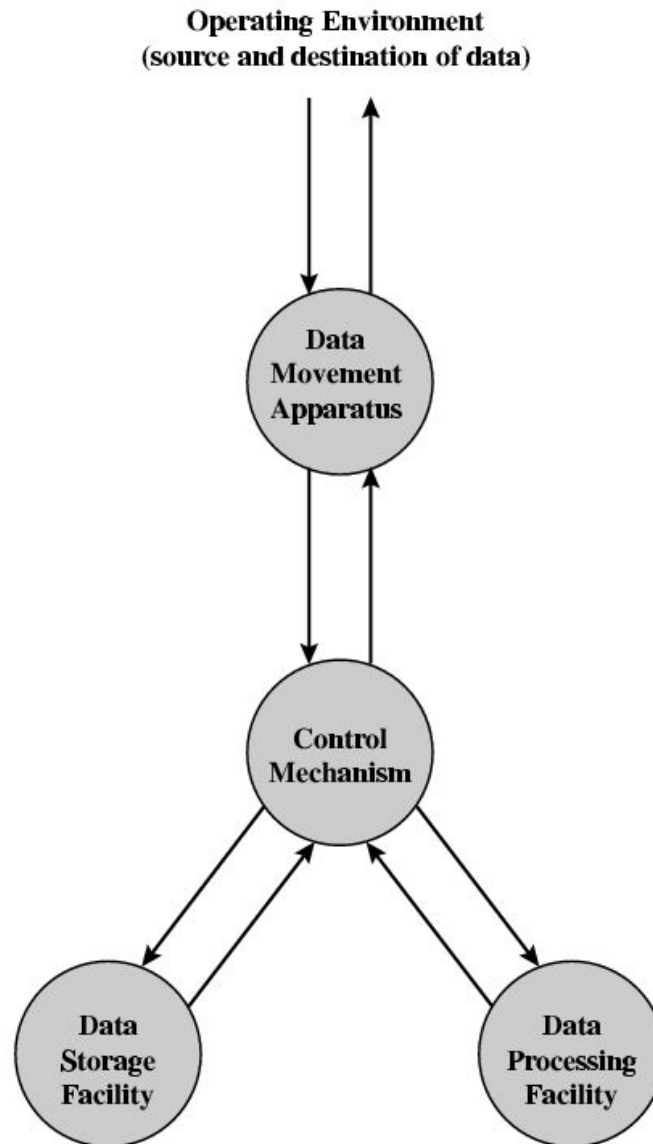
Structure & Function

- A computer is a complex system; contemporary computers contain millions of elementary electronic components.
- Structure is the way in which the components are interrelated.
- Function is the operation of individual components as part of the structure

Function

- All computer functions are:
 - Data processing
 - Data storage
 - Data movement
 - Control

Functional View



Function (1)

- The computer must be able to process data.
 - Data may take a wide variety of forms.
 - The range of processing requirements is broad.
- It is also essential that a computer store data.
 - Files of data are stored in a short-term data storage or long term-data storage on the computer for subsequent retrieval and update.

Function (2)

- The computer must be able to **move data between itself and the outside world.**
- ❖ The computer's operating environment consists of devices that serve as either **sources or destinations of data.**
- ❖ When data are received from or delivered to a device that is directly connected to the computer, the process is known as **input-output (I/O), and the device is referred to as a peripheral.**
- ❖ When data are moved over longer distances, to or from a remote devices, the process is known as **data communications.**

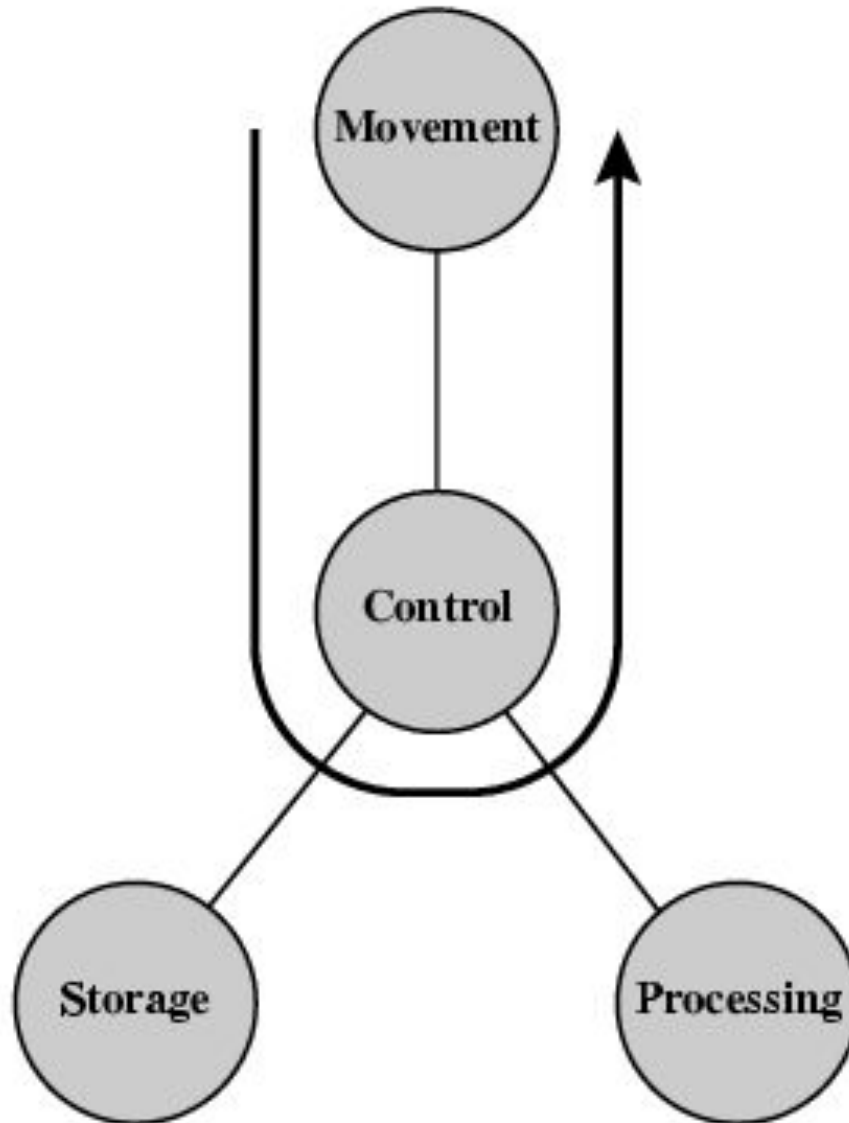
Function (3)

- Finally, there must be control of these three functions.
- This control is exercised by the individual (s) who provides the instructions to the computer.

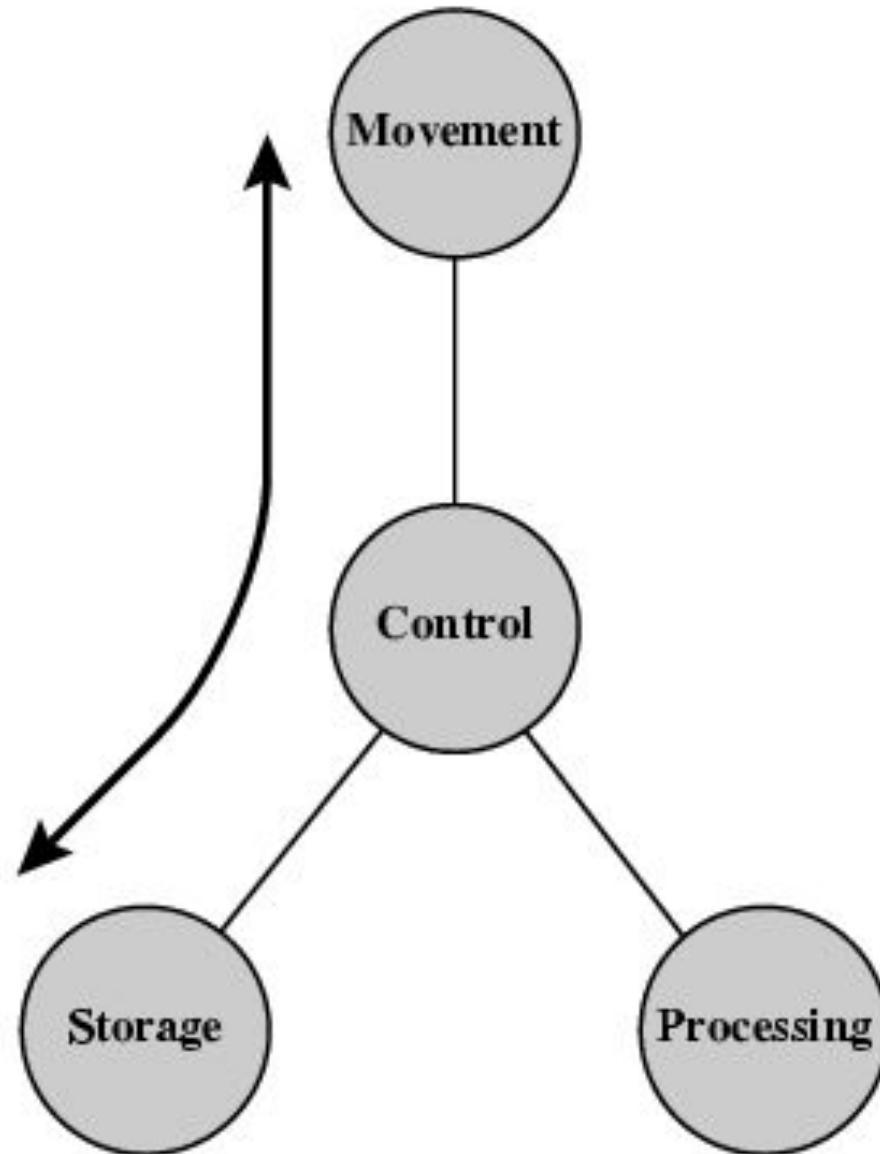
Four Possible Computer Operations

- The computer can function as a data movement device, simply transferring data from one peripheral or communications line to another [See (a)]
- It can also function as a data storage device, with data transferred from the external environment to computer storage (read) and vice versa (write) [See (b)]
- Final two operations involving data processing either from/to storage or from storage to I/O. [See (c) and (d)]

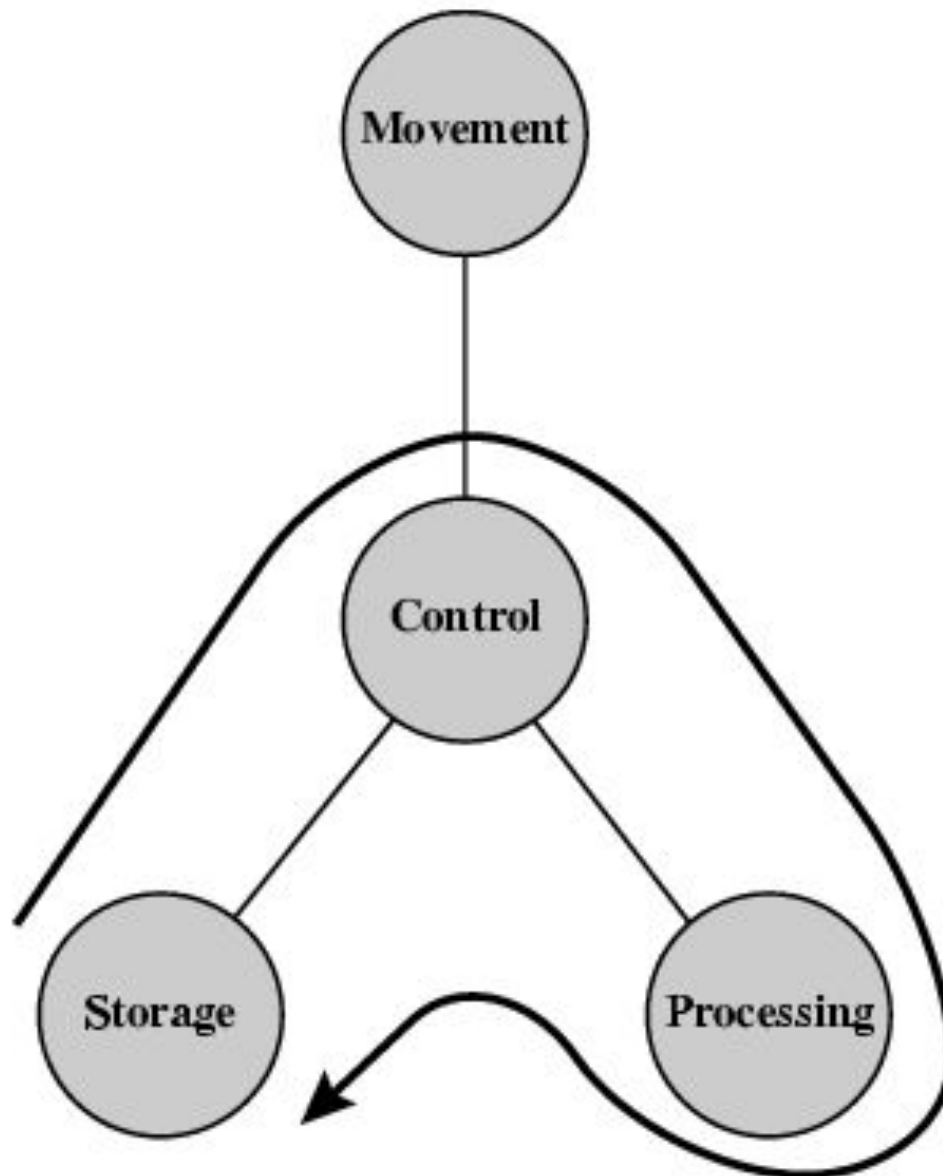
Operations (a) Data movement



Operations (b) Storage

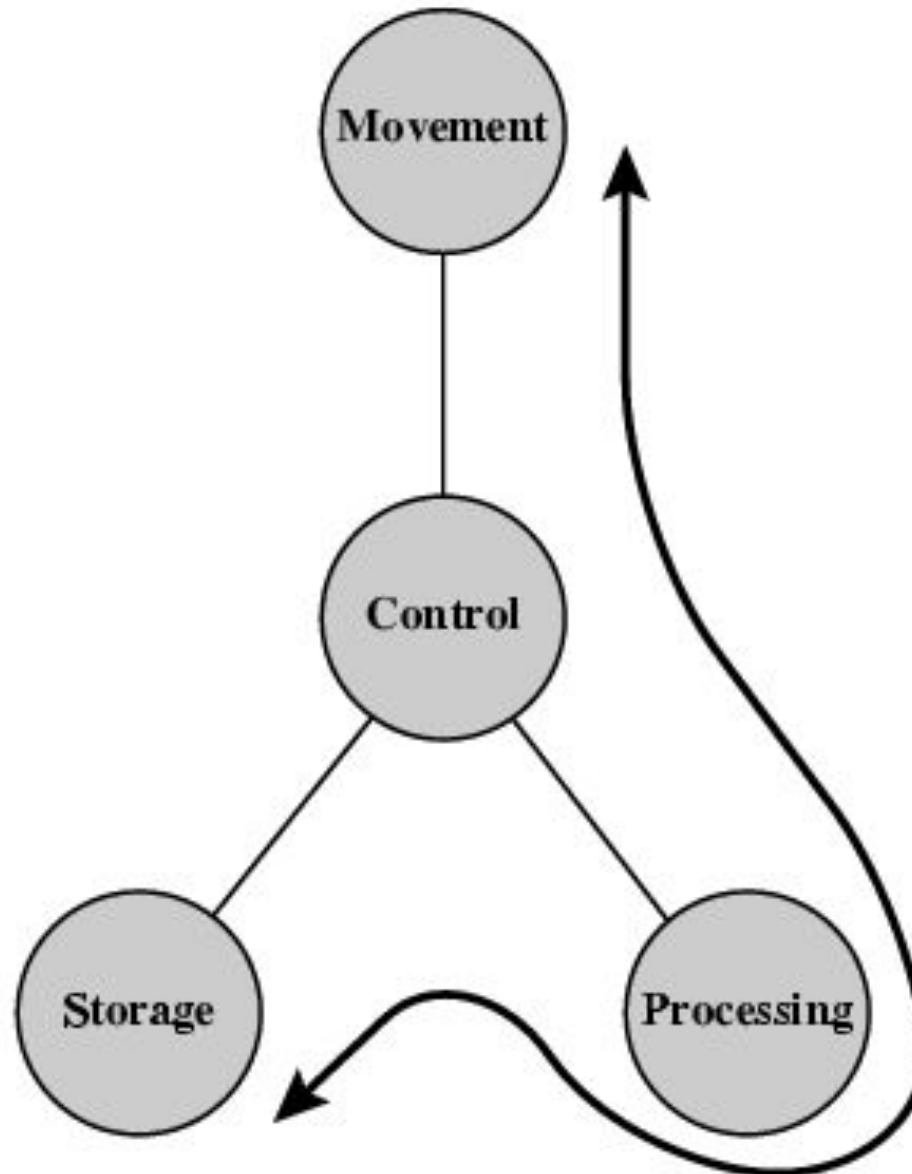


Operation (c) Processing from/to storage



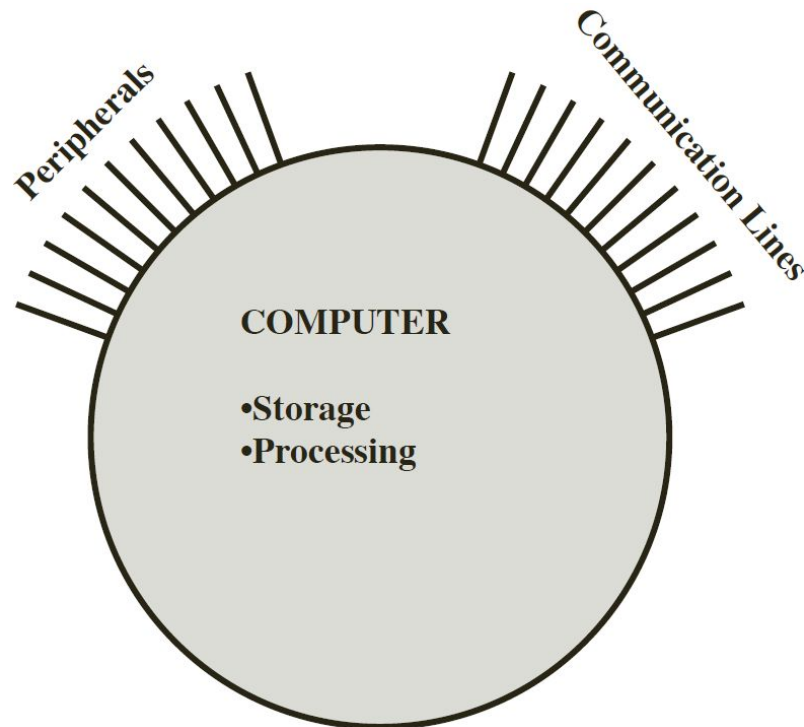
Operation (d)

Processing from storage to I/O



Structure

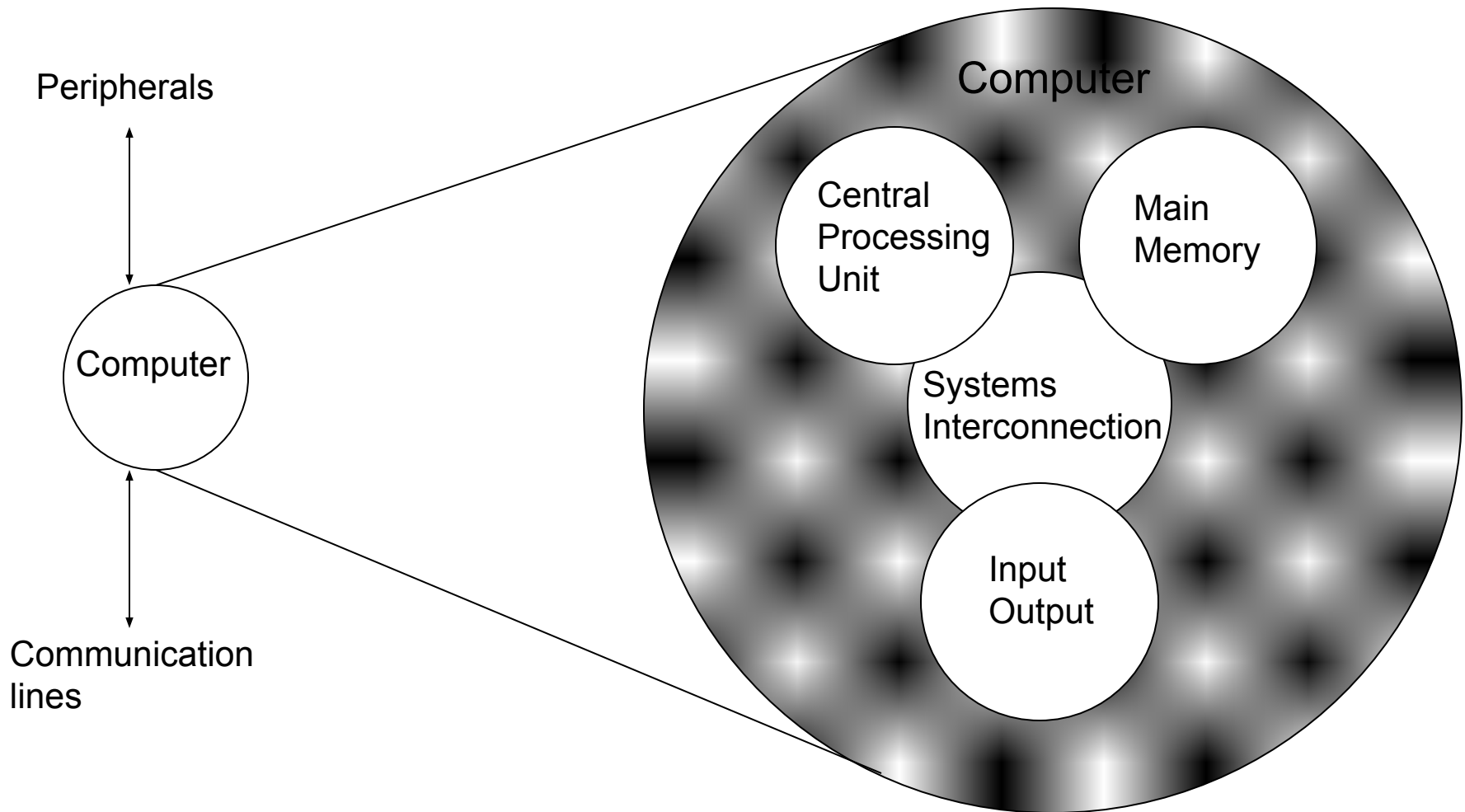
- The computer is an entity that interacts in some fashion with its external environment.
- In general, all of its linkages to the external environment can be classified as peripheral devices or communication lines.



Structural Components:

- Four main structural components:
- ❑ Central Processing Unit (CPU): Controls the operation of the computer and performs its data processing functions (referred as processor).
- ❑ Main Memory: Stores data.
- ❑ I/O: Moves data between the computer and its external environment.
- ❑ System Interconnection: Some mechanism that provides for communication among CPU, main memory, and I/O.

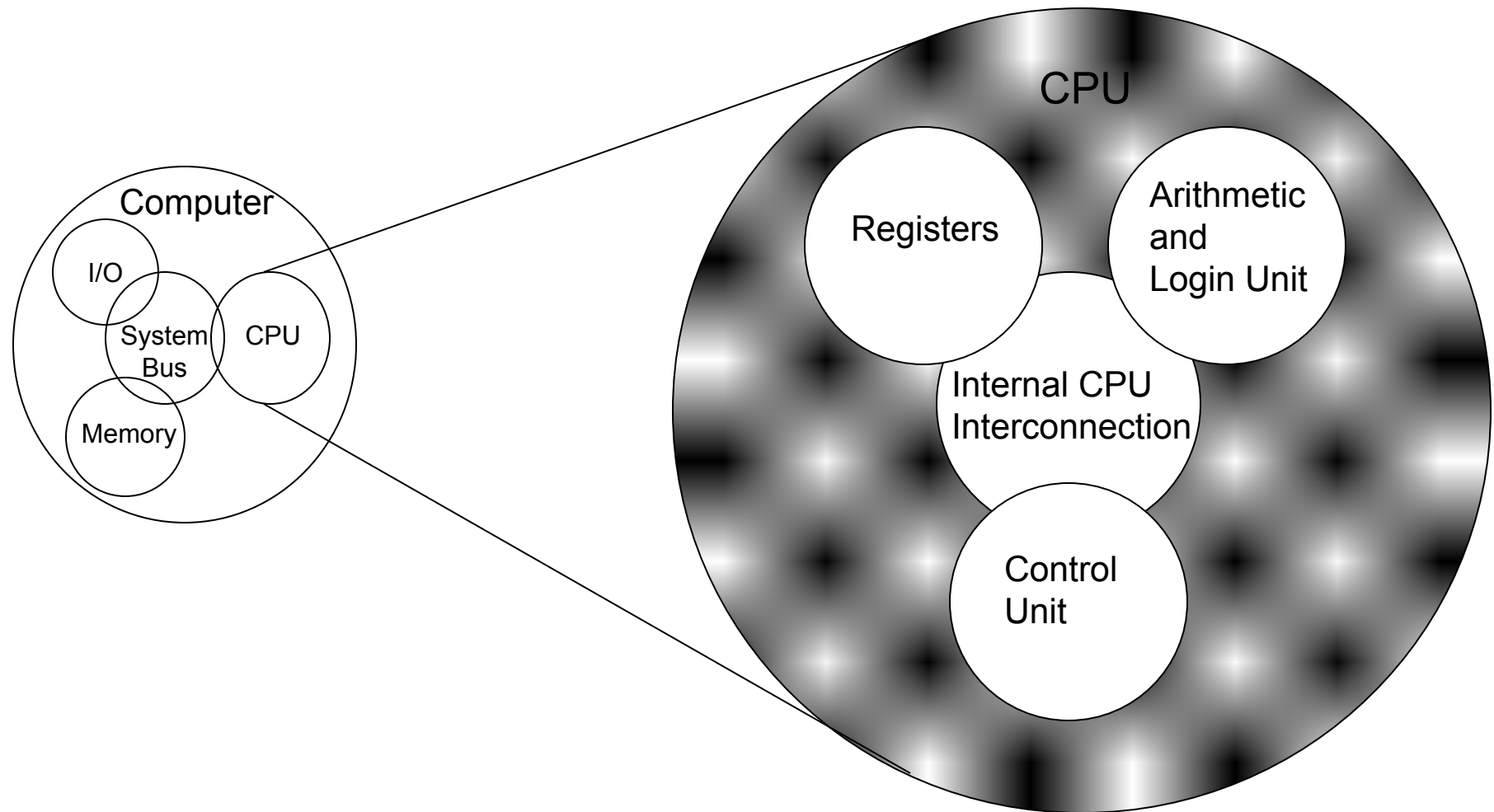
Structure - Top Level



Structure - The CPU

- Major structural components are:
- ❑ Control Unit: Controls the operation of the CPU and hence all the computer resources.
- ❑ Arithmetic and Logic Unit (ALU): Performs (Arithmetical and logical functions and controls the speed of those operations) the computer's data processing functions.
- ❑ Registers: Provide storage internal to the CPU.
- ❑ CPU Interconnection: Some mechanism that provides for communication among the control unit, ALU, and registers.

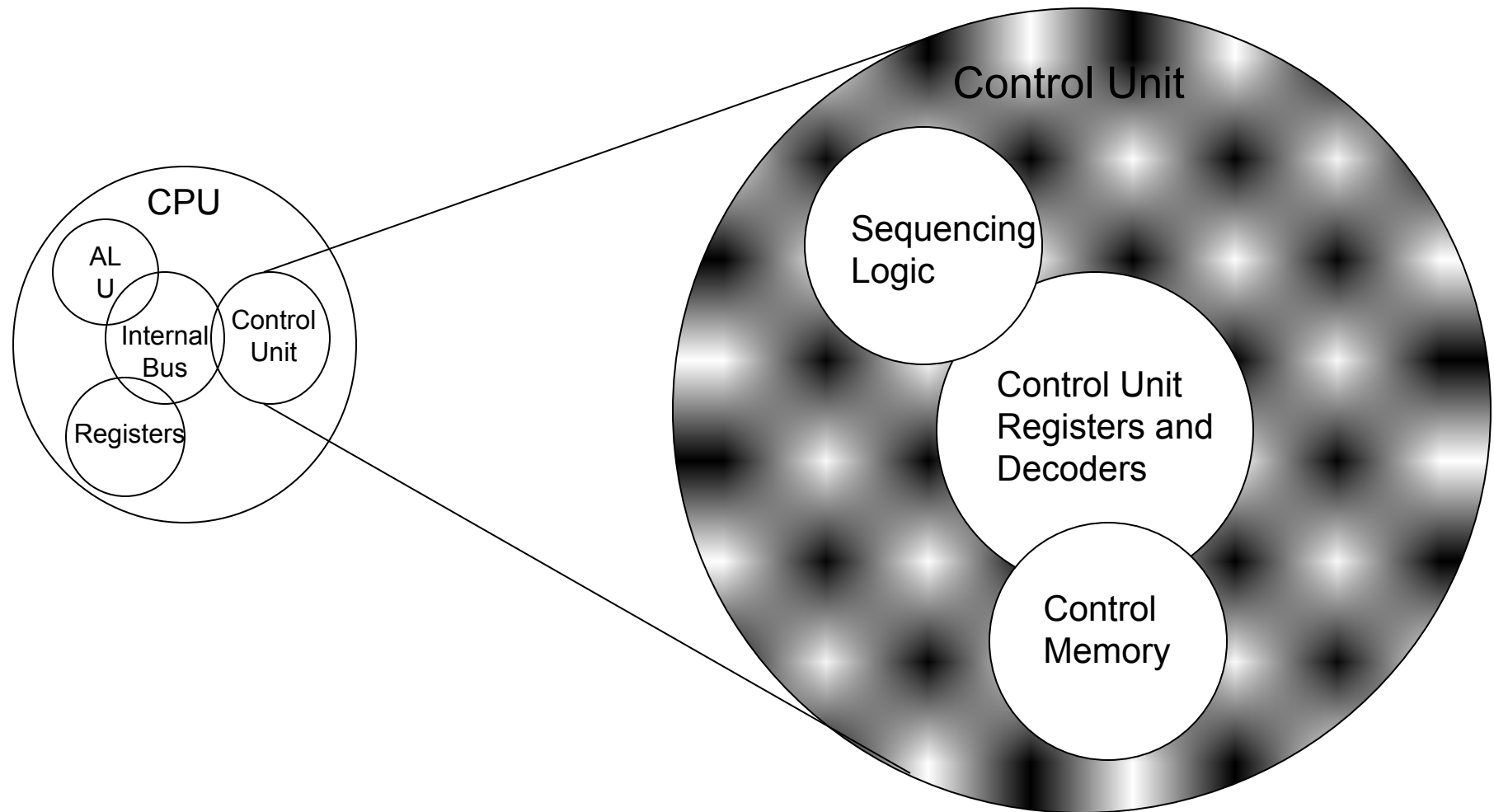
Structure - The CPU



Structure - The Control Unit

- The most interesting component is the control unit.
- The control unit can be implemented using the technique of microprogramming.
- Portion of control storage contains microcode.
- Micro-operations generally involves a transfer between registers, a transfer between a register and an external bus, or a simple ALU operation.
- It causes the processor to execute micro-operations in the proper sequence, determined by the program being executed.
- It generates the control signals that cause each micro-operation to be execute.

Structure - The Control Unit



Lecture Summary

- This lecture introduces –

- Overall course structure and overview of entire course.
- the organization and architecture of the computer.
- the structure and function of the computer.